

Nutrition Basics



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Objectives

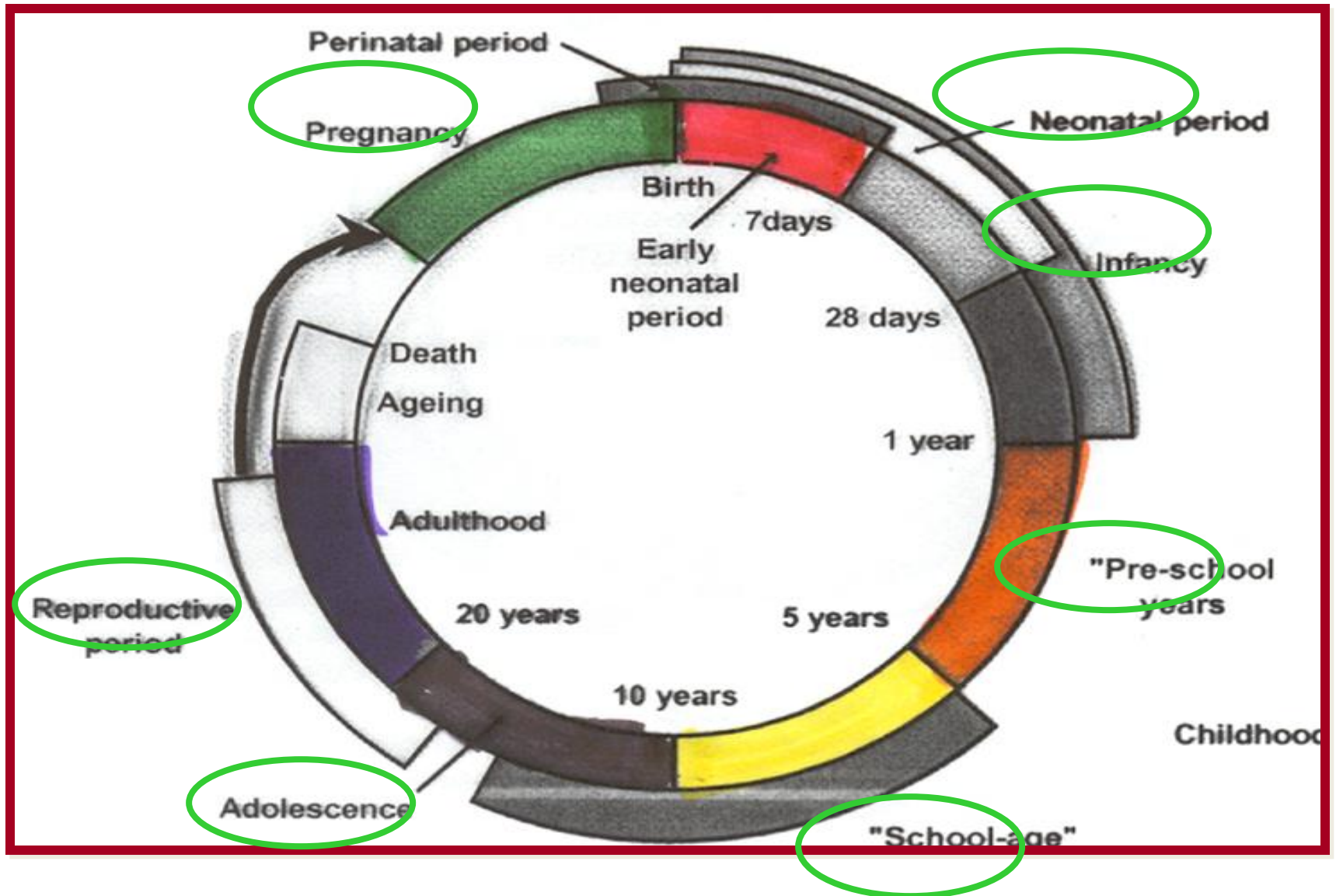
After completing this chapter students should be able to:

- ✓ *Describe the basic of principles of science of nutrition*
- ✓ *Identify nutrients and other constituents of food*
- ✓ *The life cycle approach*

Key nutrition Concepts

1. Nutrition is the study of foods, their nutrients and other chemical substances and the effect of constituents on the health.
2. Nutrition is an interdisciplinary science.
3. Nutrition recommendations for the public change as new knowledge about nutrition and health relationships is gained
4. At the core of nutrition science, there is a principles that represent **basic truths** and serve as the foundation of our understanding about the nutrition.
5. Healthy individuals require the same nutrients across the life cycle but in differing amounts.
6. Nutritional status during one stage of the life cycle influences health status during subsequent life cycle stages.

❖ Nutritional needs change at each stages of life cycle.



Principles of Human Nutrition

Table 1.1 Principles of human nutrition

Principle #1 Food is a basic need of humans.

Principle #2 Foods provide energy (calories), nutrients, and other substances needed for growth and health.

Principle #3 Health problems related to nutrition originate within cells.

Principle #4 Poor nutrition can result from both inadequate and excessive levels of nutrient intake.

Principle #5 Humans have adaptive mechanisms for managing fluctuations in food intake.

Principle #6 Malnutrition can result from poor diets and from disease states, genetic factors, or combinations of these causes.

Principle #7 Some groups of people are at higher risk of becoming inadequately nourished than others.

Principle #8 Poor nutrition can influence the development of certain chronic diseases.

Principle #9 Adequacy and balance are key characteristics of a healthy diet.

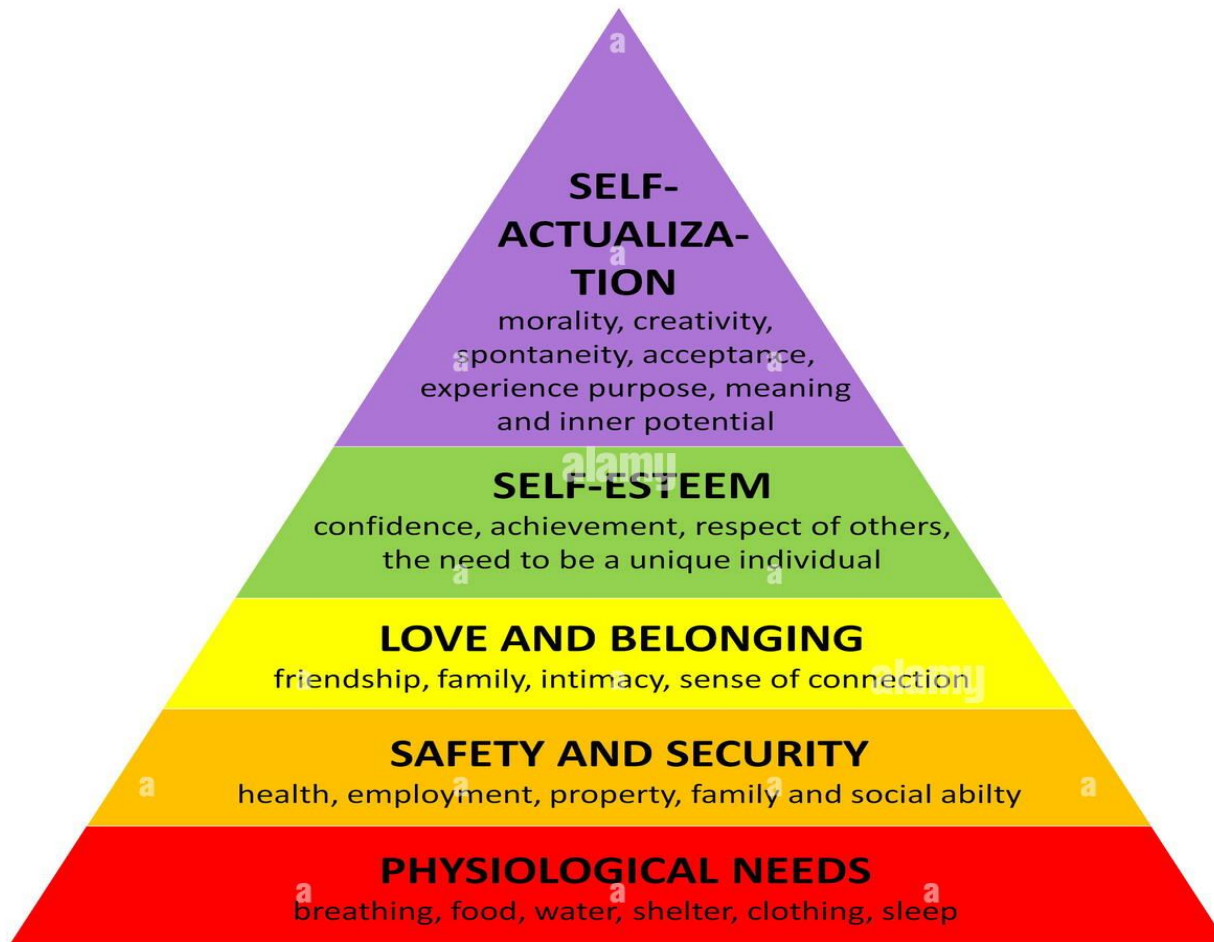
Principle #10 There are no “good” or “bad” foods.

Nutrition Principle #1

Food is a basic need of humans.

- ✓ Humans need enough food **to live** and the right assortment of foods for **optimal health**.
- ✓ People who have enough food to meet their needs at all times experience ***food security***.
 - ✓ **Food security**—having access at all times to sufficient supply of safe, nutritious foods
 - ✓ **Food insecurity**—limited or uncertain availability of safe, nutritious foods.

1.1 The need for food is part of Maslow's hierarchy of needs



Nutrition Principle #2

- Foods provide energy (calories), nutrients, and other substances needed for growth & health.
 - **Calorie**—a measure of the amount of energy transferred from food to the body
 - **Nutrients**—chemical substances in food that are used by the body

- The Six Categories of Nutrients

- Carbohydrates
- Proteins
- Fats (Lipids)
- Vitamins
- Minerals
- Water

Table 1.2 The six categories of nutrients

1. **Carbohydrates** Chemical substances in foods that consist of a single sugar molecule or multiples of sugar molecules in various forms. Sugar and fruit, starchy vegetables, and whole grain products are good dietary sources.
2. **Proteins** Chemical substances in foods that are made up of chains of amino acids. Animal products and dried beans are examples of protein sources.
3. **Fats (Lipids)** Components of food that are soluble in fat but not in water. They are more properly referred to as “lipids.” Most fats are composed of glycerol attached to three fatty acids. Oil, butter, sausage, and avocado are examples of rich sources of dietary fats.
4. **Vitamins** Thirteen specific chemical substances that perform specific functions in the body. Vitamins are present in many foods and are essential components of the diet. Vegetables, fruits, and grains are good sources of vitamins.
5. **Minerals** In the context of nutrition, minerals consist of 15 elements found in foods that perform particular functions in the body. Milk, dark, leafy vegetables, and meat are good sources of minerals.
6. **Water** An essential component of the diet provided by food and fluid.

Essential Nutrients

- Nutrients the body cannot manufacture are “essential” in the diet. We must consume them. They include:
 - Carbohydrates
 - Certain amino acids—”building blocks” of proteins
 - Essential fatty acids: linoleic acid and alpha-linolenic acid
 - Vitamins & minerals
 - Water

Nonessential Nutrients

- ✓ Present in foods, but not required in the diet since we can make them
- ✓ Examples of nonessential nutrients: cholesterol and glucose

Factors that Impact Nutrient Needs

- Age
- Body size
- Gender
- Pregnancy and lactation
- Growth
- Illness
- Medications
- Genetics

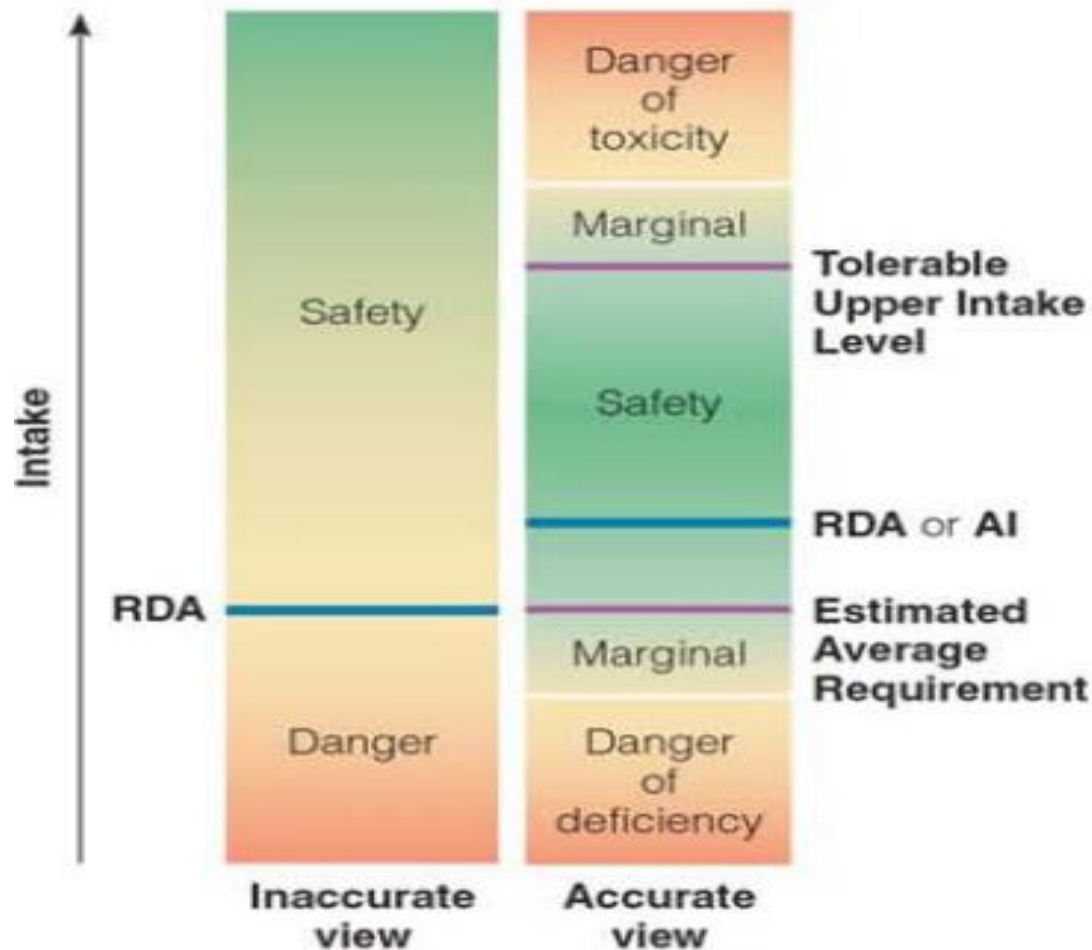
Dietary Intake Standards

- ✓ **Dietary Reference Intakes (DRIs)** general term used for the nutrient intake **standards** for healthy people.
- ✓ **Recommended Dietary Allowances (RDAs)** The average daily amount of a nutrient considered adequate to meet the known nutrient needs of practically **all healthy people/(97%-98%)**
- ✓ **Adequate Intakes (AIs):** are “**tentative**” RDAs
 - ✓ established when evidence is insufficient to develop an RDA and is set at a level assumed to ensure nutritional adequacy.

Cont....

- ✓ **Estimated Average Requirements(EARs)**- Estimated to meet the requirements of **half** the healthy individuals in a group.
- ✓ **Tolerable Upper Intake Levels (ULs)** The **maximum** daily amount of a nutrient that appears safe for most healthy people and beyond which there is an increased risk of **adverse health effects**.

1.2 Theoretical framework, terms and abbreviations used in the DRI



Question For Home Reading

- **What is the difference between Dietary guidelines and DRIs ?**
- **What is Daily Values**
- **What is the drawback of DRIs?**

Carbohydrates

- **Simple carbohydrates**
 - Monosaccharides(single molecules)
 - Disaccharides(containing two monosaccharides)
- **Complex carbohydrates**
 - Starches
 - Glycogen
 - Fiber
- **Alcohol**

Carbohydrates

- **Recommended intake level**
 - 45-65% of calories
 - Added sugar: 25% or less of calories
 - 21-35 g fiber/day for females
 - 30-38 g fiber/day for males
- **Food sources**
 - Widely distributed in plant foods
 - Milk is only animal source

Protein

- ✓ Amino acids—“building blocks” of proteins
 - ✓ Essential—body cannot make; must be provided in diet
 - ✓ Nonessential—body can make
- ✓ Protein quality—high-quality proteins provide all essential amino acids
- ✓ Recommended intake
 - 10-35% of calories

Essential and non-essential amino acids

Essential amino acids	Non essential amino acids
Lysine	<u>Glycine</u>
Tryptophan	<u>Alanine</u>
<u>Histidine</u>	Serine
<u>Phenylalanine</u>	<u>Norleucine</u>
<u>Leucine</u>	Aspartic acid
<u>Isoleucine</u>	<u>Glutamic acid</u>
<u>Threonine</u>	<u>Hydroxy-glutamic acid</u>
<u>Methionine</u>	<u>Proline</u>
<u>Valine</u>	<u>Hydroxy – proline</u>
	<u>Citrulline</u>
	Tyrosine
	<u>Cystine</u>
	<u>Arginine</u>

Food sources of protein differ in quality, based on the types and amounts of amino acids they contain.

Table 1.6 Food sources of protein

	Portion Size	Grams of Protein
Meats		
Beef, lean	3 oz	26
Tuna, in water	3 oz	24
Hamburger, lean	3 oz	24
Chicken, no skin	3 oz	24
Lamb	3 oz	22
Pork chop, lean	3 oz	20
Haddock, broiled	3 oz	19
Egg	1 med	6
Dairy Products		
Cottage cheese, low fat	½ c	14
Yogurt, low fat	1 c	13
Milk, skim	1 c	9
Milk, whole	1 c	8
Swiss cheese	1 oz	8
Cheddar cheese	1 oz	7
Grain Products		
Oatmeal, cooked	½ c	4
Pasta, cooked	½ c	4
Bread	1 slice	2
Rice, white or brown	½ c	2

Cont...

❖ Protein deficiency leads to

✓ loss of muscle tissue, growth failure, weakness, reduced resistance to disease, kidney and heart problems, and contributes to the development of a severe form of protein-energy malnutrition known as ***kwashiorkor***.

Fats (Lipids)

- Fats
 - Fats = solid at room temperature
 - Oils = liquid at room temperature
- Triglycerides— glycerol with 3 fatty acids attached

Essential Fatty Acids

- Essential fatty acids
 - Linoleic acid (omega-6)
 - Alpha-linolenic acid (omega-3)
- ✓ Play a role in normal growth & development especially in the **eyes** and **brain**
- ✓ Prevent *heart* disease, *hypertension*, *arthritis* & *cancer*

Cont...

- Fats and oils are a concentrated source of energy, providing **9 calories** per gram.
- Fats perform a number of important functions in the body.

They are:

- precursors for cholesterol and sex-hormone synthesis,
- components of cell membranes,
- vehicles for carrying certain vitamins that are soluble in fats only, and
- suppliers of the *essential fatty acids* required for growth and health.

Saturation States

- Saturated—no double bonds
- Unsaturated
 - Monounsaturated—one double bond
 - Polyunsaturated—2 or more double bonds
- Hydrogenation and *Trans* Fats
 - Hydrogenation—adds hydrogen to unsaturated fatty acids
 - Changes structure of fatty acid from *cis* to *trans* structure

Key:

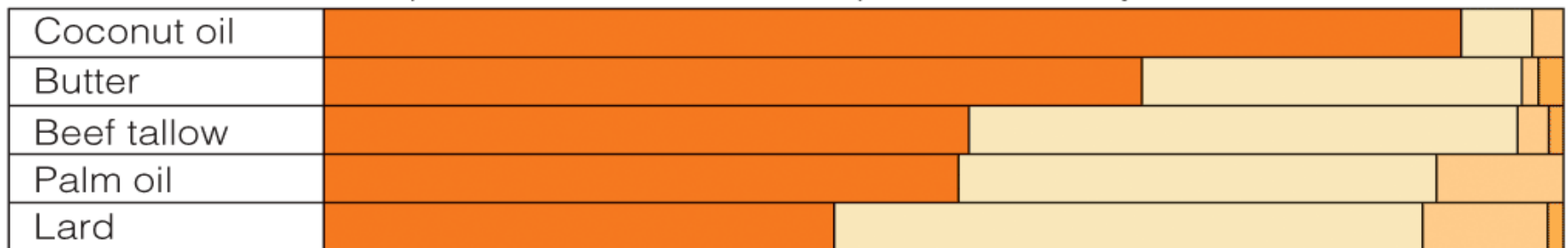
Saturated fats

Polyunsaturated,
omega-6 fats

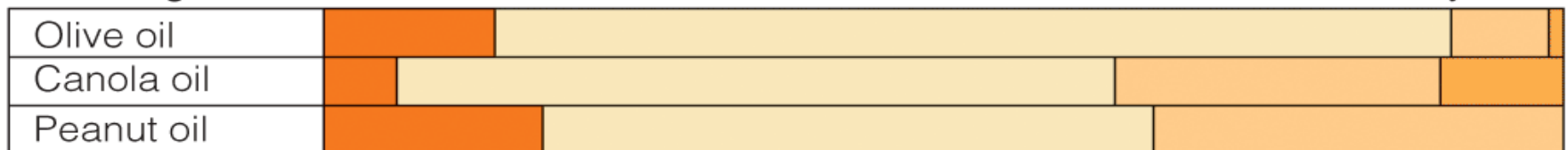
Monounsaturated fats

Polyunsaturated,
omega-3 fats

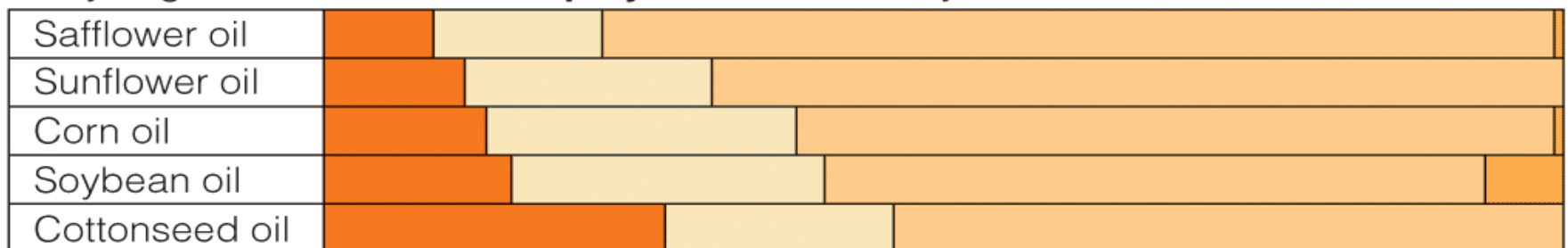
Animal fats and the tropical oils of coconut and palm are mostly **saturated**.



Some vegetable oils, such as olive and canola, are rich in **monounsaturated** fatty acids.



Many vegetable oils are rich in **polyunsaturated** fatty acids.



Cholesterol

- Cholesterol
 - A fat-soluble, clear liquid found in animal products
 - component of all animal cell membranes, the brain, and the nerves
 - A precursor of estrogen, testosterone, and vitamin D
 - There is no recommended level of cholesterol intake
 - The body is able to produce enough cholesterol

Recommended Intake of Fats

- Not all fats are created equal
- “Unhealthy” fats
 - Those that raise LDL-cholesterol
 - Examples: *Trans* and saturated fats and cholesterol
- “Healthy” fats
 - Those that raise HDL-cholesterol
 - Examples: Monounsaturated, polyunsaturated, linolenic,

Recommended Intake of Fats

- ❑ 20-35% of calories from fat, limiting unhealthful fats as much as possible
- **Linoleic acid**: 17 g/day for men, 12 g/day for women
- **Alpha-linolenic acid**: 1.6 g/day for men, 1.1 g/day for women
- Sources of fats :milk and milk products, meat, peanut, chicken..

Vitamins

- ✓ Vitamins are the micro-nutrients required for the proper growth and maintenance of our body and organs.
- ✓ They help with the **metabolism of proteins, fats and carbohydrates** required for the body development.
- ✓ Vitamins are **not developed/synthesized** inside human body.
- ✓ Hence, our food should have appropriate amounts of vitamins meet the required levels.

Types

- Water-soluble vitamins
 - thiamin, riboflavin, niacin, B₆, folate, B₁₂, biotin, pantothenic acid, C
- Fat-soluble vitamins: Fat-soluble vitamins are stored in the body's fat tissues and the liver.
 - A, D, E, K

Cont...

❖ Excessive consumption of the fat-soluble vitamins from supplements, especially of vitamins A and D, produces various symptoms of toxicity.

- Functions

- coenzymes
- antioxidants

Vitamin Solubility

Table 1.8 Vitamin solubility

Water-Soluble Vitamins	Fat-Soluble Vitamins
B-complex vitamins	Vitamin A (retinol, beta-carotene)
Thiamin (B ₁)	Vitamin D (1,25 dihydroxy- cholecalciferol)
Riboflavin (B ₂)	Vitamin E (alpha-tocopherol)
Niacin (B ₃)	Vitamin K
Vitamin B ₆	
Folate	
Vitamin B ₁₂	
Biotin	
Pantothenic acid	
Vitamin C (ascorbic acid)	

The Many Roles of Vitamins in Promoting Health

Metabolic Function	Vitamins That Play a Role
Antioxidants	Vitamin C, vitamin E
Blood clotting and red blood cell synthesis	Folate, vitamin B ₆ , vitamin B ₁₂ , vitamin K
Bone health	Vitamin A, vitamin C, vitamin D, vitamin K
Energy	Biotin, niacin (B ₃), pantothenic acid, riboflavin (B ₂), thiamin (B ₁), vitamin B ₆ , vitamin B ₁₂
Growth and reproduction	Vitamin A, vitamin D
Immune function	Vitamin A, vitamin C, vitamin D
Protein metabolism	Folate, vitamin B ₆ , vitamin B ₁₂

Vitamins

- Read your Reference for the following.
- ✓ Recommended intakes and food sources of each Vitamins
- ✓ Deficiencies
- ✓ Toxicities(Excessive consumption) of each.

Other substances in food

- ✓ **Phytochemicals:** Chemical substances in plants.
- ✓ High intakes of certain phytochemicals may partially account for lower rates of heart disease and cancer
- ✓ **Antioxidants:** Chemical substances that prevent or repair damage to cells caused by exposure to oxidizing agents.
- ✓ Many different antioxidants are found in foods; some are made by the body.

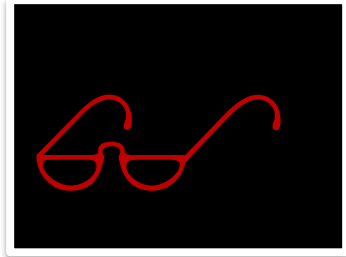
Minerals

There are 15 essential minerals

Table 1.11 Minerals required by humans

Calcium	Fluoride	Chromium
Phosphorus	Iodine	Molybdenum
Magnesium	Selenium	Sodium
Iron	Copper	Potassium
Zinc	Manganese	Chloride

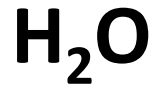
Minerals



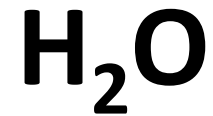
- Minerals are unlike other nutrients in that they consist of single atoms and carry a charge in solution.
- All 15 minerals vary in:
 - Functions
 - Deficiencies
 - Overdoses

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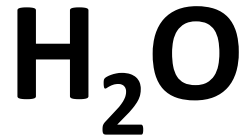
- The charge carried by minerals allows them to combine with other minerals to form stable complexes.
- In body fluids, charged minerals serve as a **source of electrical power** that stimulates muscles to contract (e.g., the heart to beat) and nerves to react.
- Minerals also help the body maintain an **adequate amount of water** in tissues and **control** how acidic or basic body fluids remain.



Water



- Adults are 60-70% water
- Recommended daily intakes
 - 12-16 cups for males
 - 11 cups for females
 - 75% from fluids; 25% from foods
- Dietary sources
 - Best to drink water & nonalcoholic beverages
 - Alcoholic beverages increase water loss through urine



Cont...

It plays a role in:

- Energy transformation,
- Excretion of wastes, and
- Temperature regulation

Nutrition Principle #3

- **Health problems related to nutrition originate within cells.**
- The functions of each cell are maintained by the nutrients it receives.
- Problems arise when a cell's need for nutrients differs from the amounts that are available.

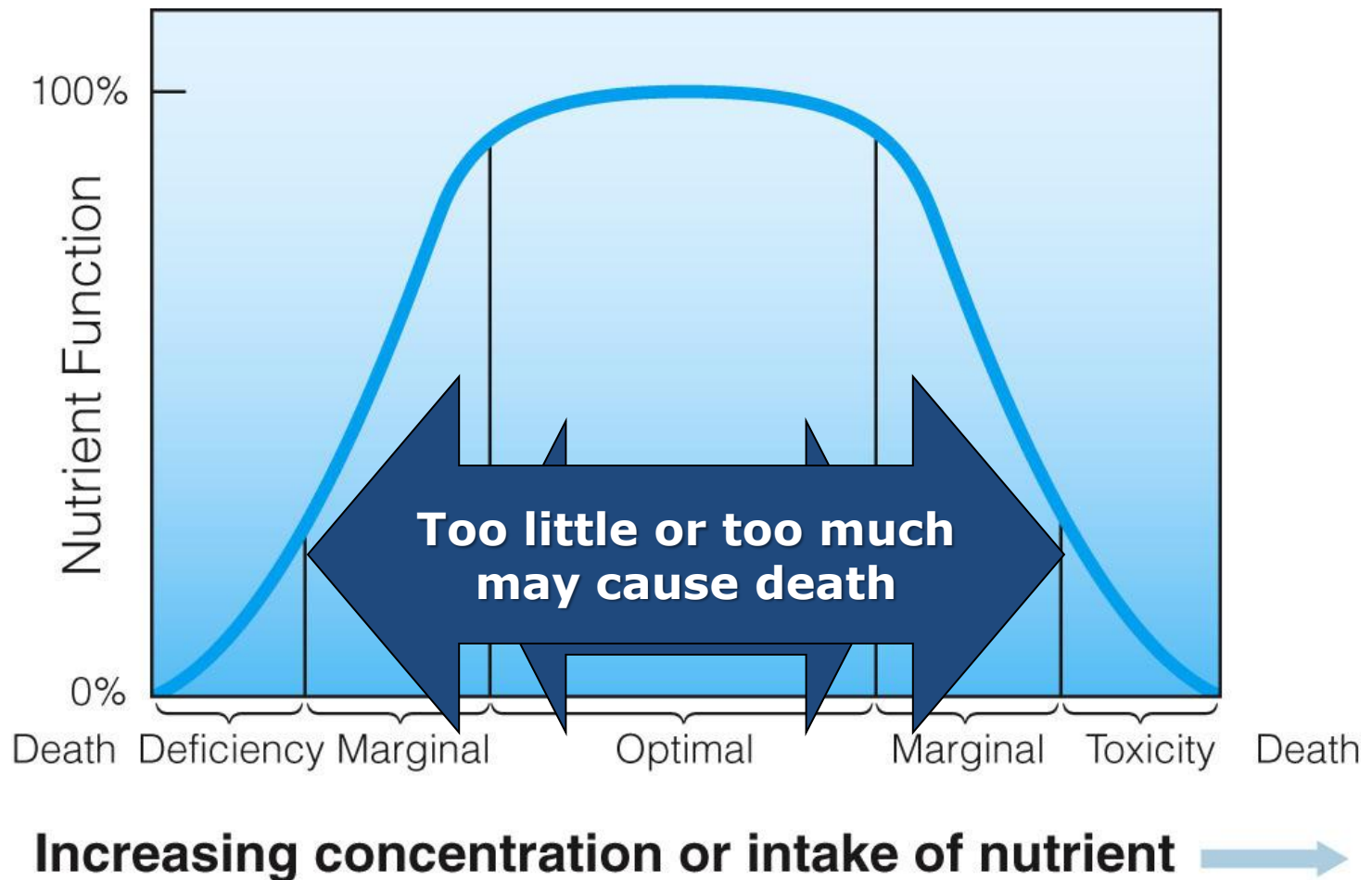
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- Disruptions in the availability of nutrients, or the presence of harmful substances in the cell's, initiate diseases and disorders.
- For example, folate, a B vitamin, is required for protein synthesis within cells.

Nutrition Principle #4

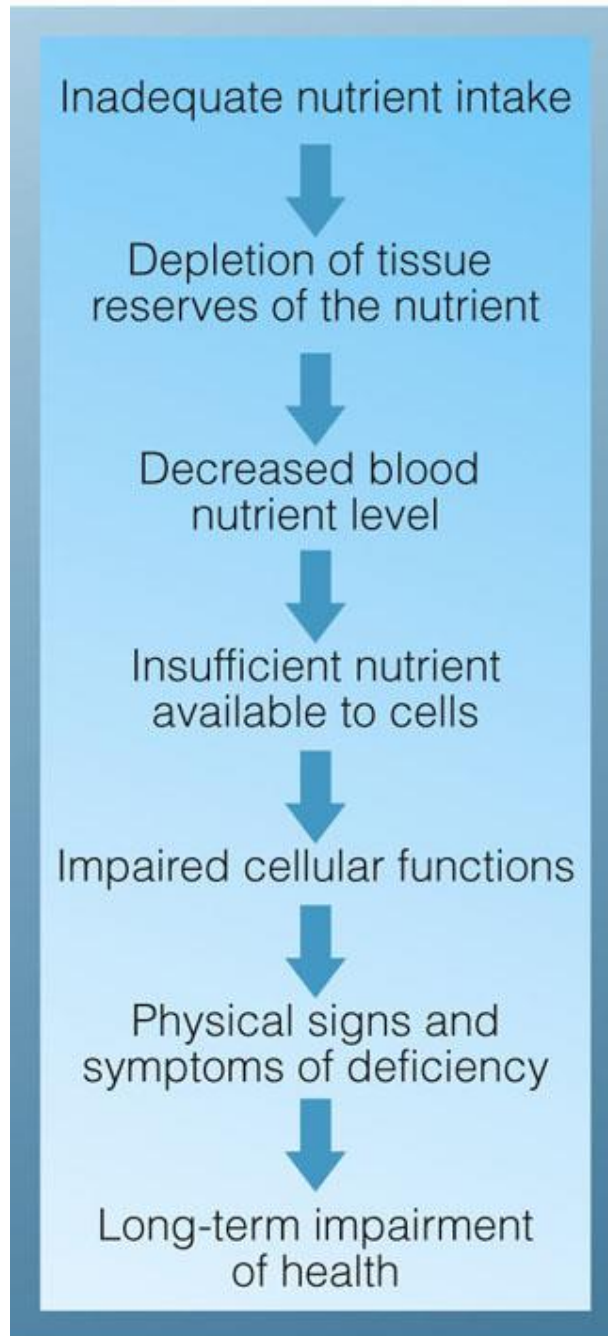
- **Poor nutrition can result from both inadequate and excessive levels of nutrient intake.**
 - Prolonged inadequate intake results in obvious deficiencies.
 - Overdoses of nutrients (usually by supplements) result in mild to severe alterations in functioning.

Nutrient Function and Consequences by Level of Intake

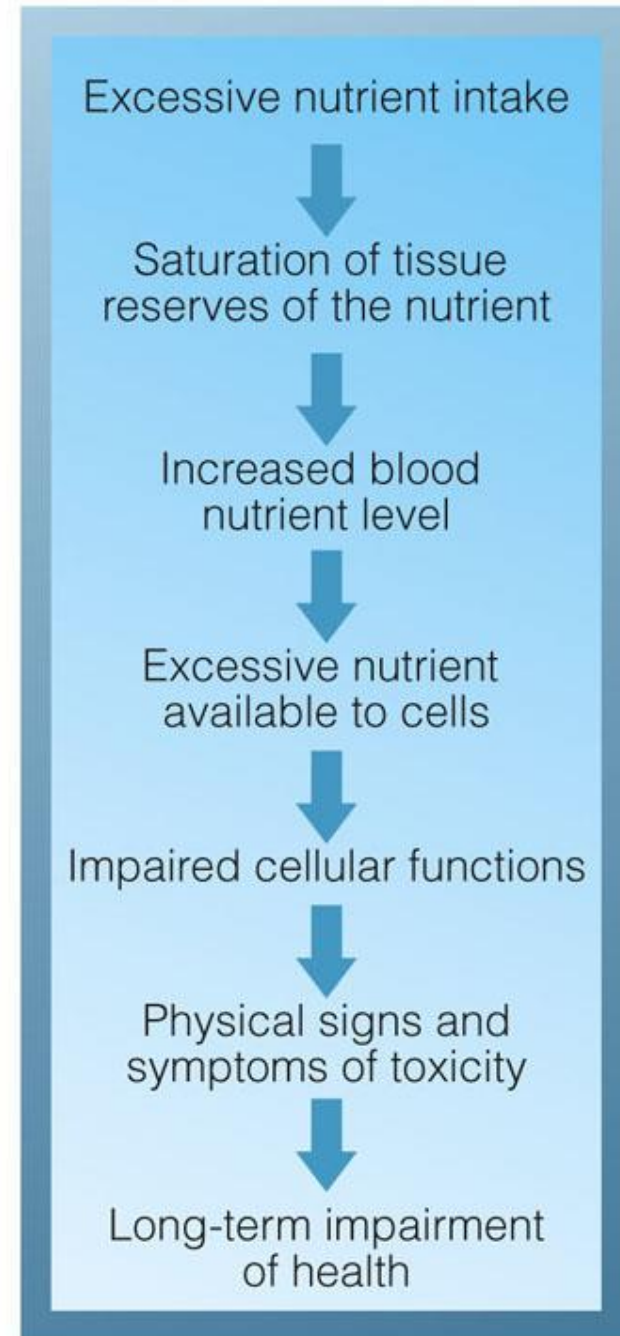


Steps in Development of Nutrient Deficiencies and Toxicities

Deficiency



Toxicity



Nutrient Deficiencies are Usually Multiple

- ✓ Most foods contain many nutrients, so poor diets are generally inadequate in many nutrients.
- ✓ Calcium and vitamin D, for example, are present in milk.

Nutrition Principle #5

- Humans have adaptive mechanisms for managing fluctuations in food intake.
 - **nutrient storage, regulation of absorption, regulation of appetite**

Nutrition Principle #6

- Malnutrition can result **from poor diets and from disease states, genetic factors, or combinations of these causes.**
 - primary malnutrition—dietary in origin
 - secondary malnutrition—precipitated by a disease state, surgical procedure, or medication

Nutrition Principle #7

- ❖ Some groups of people are at higher risk of becoming inadequately nourished than others.
 - ✓ pregnant/breastfeeding women, infants, children, people who are ill, frail elderly persons

Nutrition Principle #8

- ✓ Poor nutrition can influence the development of certain chronic diseases:

Heart disease, hypertension, cancer, stroke, osteoporosis, type 2 diabetes, obesity

Noncommunicable Diseases (2006-2015)

	2005		2006-2015 (cumulative)		
Geographical regions (WHO classification)	Total deaths (millions)	NCD deaths (millions)	NCD deaths (millions)	Trend: Death from infectious disease	Trend: Death from NCD
Africa	10.8	2.5	28	+6%	+27%
Americas	6.2	4.8	53	-8%	+17%
Eastern Mediterranean	4.3	2.2	25	-10%	+25%
Europe	9.8	8.5	88	+7%	+4%
South-East Asia	14.7	8.0	89	-16%	+21%
Western Pacific	12.4	9.7	105	+1	+20%
	58.2	35.7	388	-3%	+17%

(WHO, Chronic Disease Report, 2005)

WHO projects that over 10 years, the largest increase in deaths from cardiovascular disease, cancer, respiratory disease and diabetes will occur in developing countries.

Noncommunicable Diseases

4 Diseases, 4 Modifiable Shared Risk Factors

	Tobacco Use	Unhealthy diets	Physical Inactivity	Harmful Use of Alcohol
Cardio-vascular				
Diabetes				
Cancer				
Chronic Respiratory				

Nutrition Principle #9

- Adequacy, variety and balance are key characteristics of a healthy diet.
- Variety is a core characteristic of healthy diets because the essential nutrient and phytochemical content of foods differ.
 - Nutrient density
 - “empty-calorie” foods

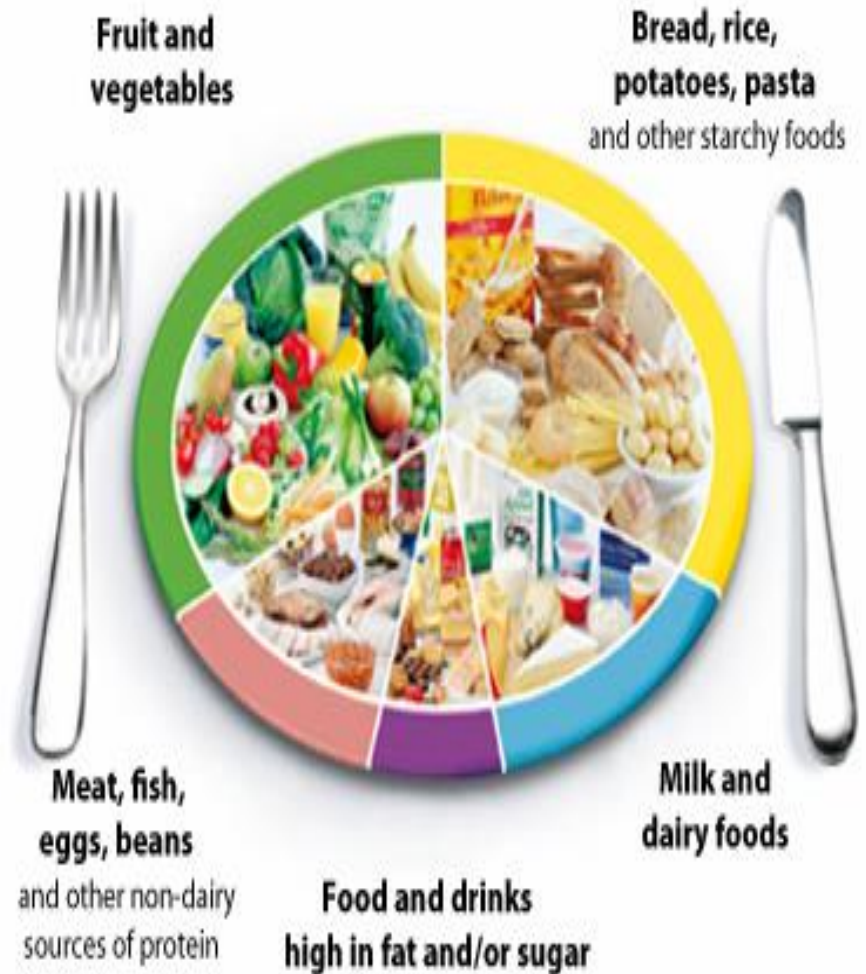
To plan a balanced diet one has to think about two things:

The mixture of foods

and

The amount of food a

person eats.



YOU ARE WHAT



YOU EAT

YOU ARE
WHAT YOU EAT



Nutrition Principle #10

- There are no “good” or “bad” foods.
 - As Hippocrates expressed it, “All things in nutriment are good or bad relatively.”
- **Nutrient-Dense Foods**—contain high amounts of nutrients to calories
- **Empty-Calorie Foods**—contain low amounts of nutrients to calories
- All foods can fit into a healthful diet as long as nutrients needs are met at calorie-intake levels that maintain a healthy body weight.

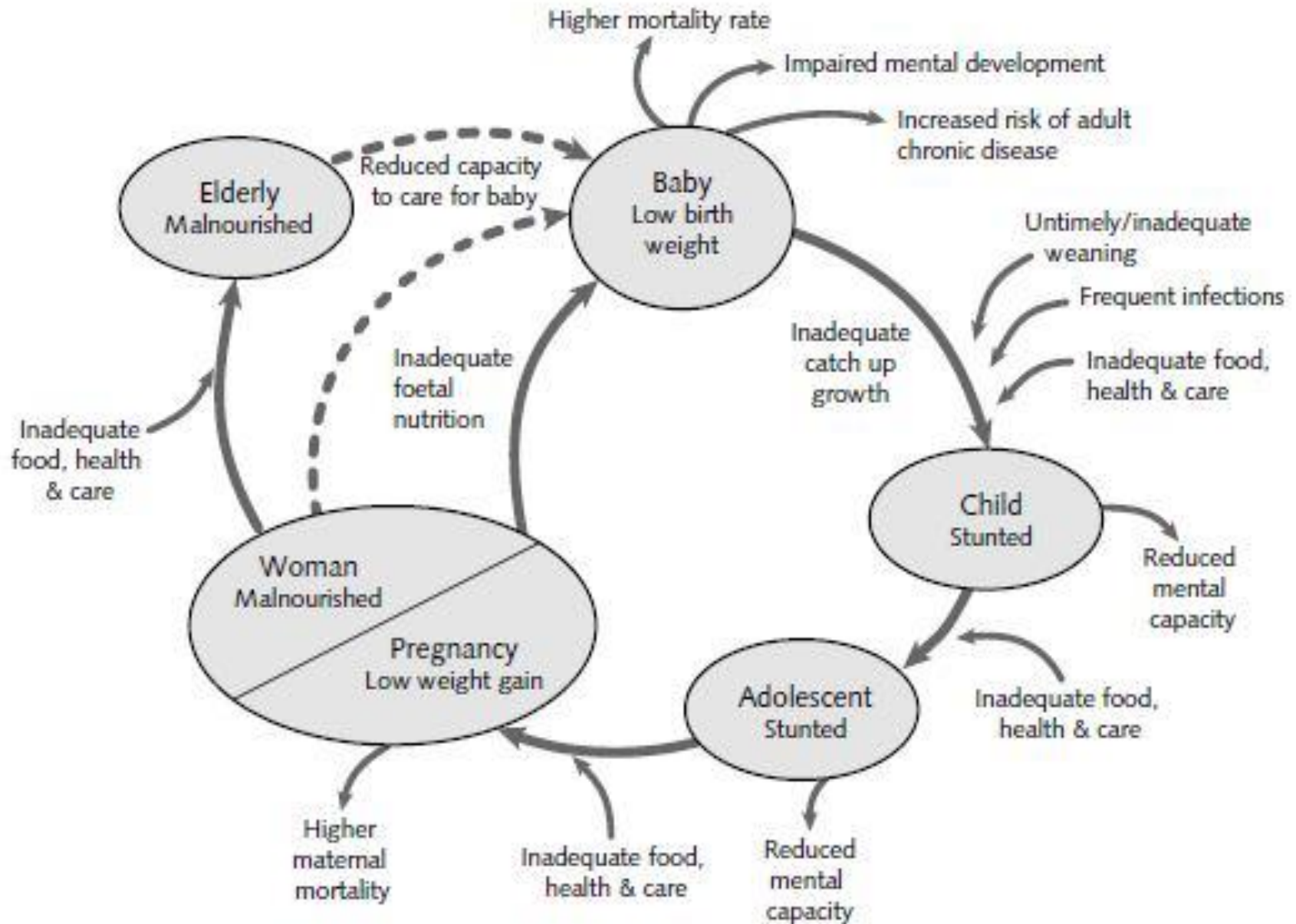
Lifecycle approach to nutrition

- ✓ The life cycle provides a strong framework for discussing the challenges facing human nutrition
- ✓ Poor nutrition starts in **utero** and extends throughout the lifecycle
- ✓ Nutritional needs should be met at **every stage of the life cycle** because nutritional **status at one stage influences health status in the next ones**.
- ✓ A stunted girl is most likely to become a stunted adolescent and later a stunted woman.

Lifecycle approach to nutrition ...

- ✓ Apart from direct effects on her health and productivity, adult stunting and underweight increase the chance that her children will be born with low birth weight.
- ✓ Nutrition interventions in pregnancy and early childhood can result in improvements in body size and composition in adolescents and young adults.
- ✓ Investing in maternal and childhood nutrition will have both short and long term benefits of huge economic and social significance.

Lifecycle: the proposed causal links...



Thank you